# MOHANRAJ A

LinkedIn | GitHub | Portfolio | LeetCode

mohn08052006@gmail.com | +91 8248777476 | Tamilnadu, India

## SKILLS

- **Languages**: Python (Advanced), JavaScript (ES6+), C++, PHP, HTML5, CSS3, SQL.
- **Frameworks & Libraries**: React.js, Next.js, Node.js, Express.js, Redux, Tailwind CSS,Django.
- **Databases**: MongoDB, MySQL, PostgreSQL, Cloud SQL, DBMS.
- **DevOps & Tools**: Git, GitHub, Docker, Kubernetes, Google Cloud Platform (GCP), GitHub Actions, JSON, REST API, Postman, Wireshark.
- **Methodologies**: Agile, Software Development Life Cycle (SDLC), CI/CD Automation.

## PERSONAL STATEMENT

Backend-focused Software Engineer with strong foundations in data structures, algorithms, and low-level memory concepts. Experienced in designing scalable REST APIs, optimizing database queries, and building efficient backend systems using Python, Rust, and Node.js. Passionate about writing performance-driven code, improving system throughput, and minimizing memory overhead. Hands-on experience with containerized deployments, cloud infrastructure on Google Cloud Platform, and CI/CD automation. Actively practice competitive programming and enjoy solving complex algorithmic problems that enhance computational efficiency. Interested in distributed systems, backend architecture, and building reliable, high-performance services that scale effectively.

## EXPERIENCE

### Software Engineer Intern | KalviumLabs | Letter

- • Designed and implemented end-to-end CI/CD pipelines using GitHub Actions, enabling automated build, test, and deployment of 3 full-stack applications to Google Cloud Platform with zero manual intervention.
- • Provisioned and managed GCP infrastructure including Cloud Run, Cloud SQL, Cloud Jobs, and Compute Engine (VMs) to support scalable, production-ready environments.
- • Containerized applications using Docker and deployed them using Kubernetes, improving deployment consistency and rollback reliability.
- • Diagnosed and resolved critical deployment failures that caused service outages, restoring system availability and implementing preventive automation to avoid recurrence.
- • Collaborated with developers to integrate automated testing and deployment workflows, reducing release cycle time and improving deployment confidence.

## PROJECTS

### Kubernetes Homelab Monitoring Stack | Grafana, Prometheus

- Built and managed a self-hosted Kubernetes homelab to deploy and monitor containerized workloads.
- Implemented Prometheus-based metrics collection and Grafana dashboards for real-time cluster observability.
- Monitored node health, pod performance, and resource utilization using exporters and custom alerts.

### Kaasu - AI-Powered Financial Advisor (MERN Stack) | Github | LIVE

- Developed a full-stack financial management platform using MongoDB, Express.js, React, and Node.js, facilitating real-time expense tracking for families.
- Integrated AI & OCR APIs to automate transaction analysis from PDF statements, eliminating manual entry and enhancing data accuracy.
- Implemented WebSockets for real-time collaboration and Redux for state management to handle complex family budget data across multiple users.
- Secured user data and file uploads using JWT authentication and Multer, and deployed the solution to Vercel and Render.

**EDUCATION**: B.Tech, Software Product Engineering
Kalvium UG Program in CS — Lovely Professional University, Punjab (2024–2028)

**ACHIEVEMENTS**

- • **Sponsored Open-Source Contributor**: Recognized by TSCircuit (1.5k+ GitHub stars) for meaningful contributions to core components.
- • **Winner, Inter-College Hackathon**: Secured 1st place and ₹10,000 for designing and pitching a Django-based web application.
- • **Competitive Programmer**: Solved 150+ algorithmic problems on LeetCode using C++ and Python; ranked consistently in the top 20%.